

PREDICTIVE MEMORY LOCALIZATION: FORECASTING INTERVENTION PATHS FOR SELECTIVE CONTROL

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ABSTRACT

Activation steering turns localized representations into control directions, but localization alone does not reveal whether a direction has a selective operating regime. We introduce *Predictive Memory Localization* (PML), which treats the measured-grid intervention path as the predictive object of memory localization. PML separates random-calibrated target movement from semantic-neighbor and capability damage, and compares static localization and supervised geometry with a strength-disjoint low-dose causal response. Our frozen study covers 3,000 records from nine datasets and fourteen domains, yielding 30,000 distinct record-direction-layer paths and 210,000 distinct path-strength evaluations. At layer 7, the geometry-derived RFM/AGOP direction reaches 13.1% target-any and 12.3% clean-any, exceeding random by 3.6 and 3.4 percentage points under a record-paired bootstrap. Across record-, dataset-, and domain-grouped splits, responses at $|\alpha| = 0.1$ are the strongest signal for outcomes at disjoint strengths $|\alpha| = 0.25$ or 0.5. On held-out records, a predictor-driven selector chooses a coefficient or abstains, improves utility and reduces semantic-neighbor damage relative to a train-tuned fixed-strength policy, and avoids most evaluations in a dense scan. Across three residual-norm-matched base models, learned directions retain selective-path gains and low-dose responses yield 0.801–0.828 record-held-out macro AUROC. PML therefore turns memory localization into a falsifiable forecast of margin-level selective outcomes and a risk-aware intervention decision.

1 INTRODUCTION

Language-model internals have been associated with feed-forward memories, knowledge neurons, and causally important states (Geva et al., 2021; Dai et al., 2022; Meng et al., 2022). This literature suggests an operational hypothesis: once information is localized, the corresponding representation should provide a useful control point. Activation steering tests that hypothesis without changing model parameters (Turner et al., 2024; Rinsky et al., 2024; Zou et al., 2025; Singh et al., 2024). Yet behavior varies across prompts, layers, models, and coefficients, and stronger interventions can damage specificity or unrelated capabilities (Li et al., 2023; Stoehr et al., 2024; Tan et al., 2024; Goyal & Daumé III, 2026). Steering is thus a multidimensional control problem (Wu et al., 2025; Pres et al., 2024; Xu et al., 2026b).

Localization describes a representation, whereas controllability is revealed across a *measured intervention path*. Target leverage can coexist with semantic-neighbor or capability damage, and a clean effect at one coefficient need not persist at another. Predicting calibrated target, damage, and clean outcomes over a pre-specified grid asks which localized directions admit a usable measured coefficient. A single endpoint cannot answer this question: the same direction may show leverage at one measured strength, damage at another, or no coefficient where the two separate cleanly.

We introduce *Predictive Memory Localization* (PML), which maps each record, direction, and layer to those random-calibrated outcomes. It organizes predictive evidence from baseline belief and metadata, static localization, and supervised representation geometry to a strength-disjoint low-dose causal response. This design directly tests whether static evidence forecasts later behavior or whether an inexpensive causal measurement is required. A collateral-aware policy then selects a coefficient or abstains. Unlike generation-level concept prediction (Fan et al., 2026), PML audits localized factual and reasoning directions with explicit collateral probes and strength-disjoint labels.

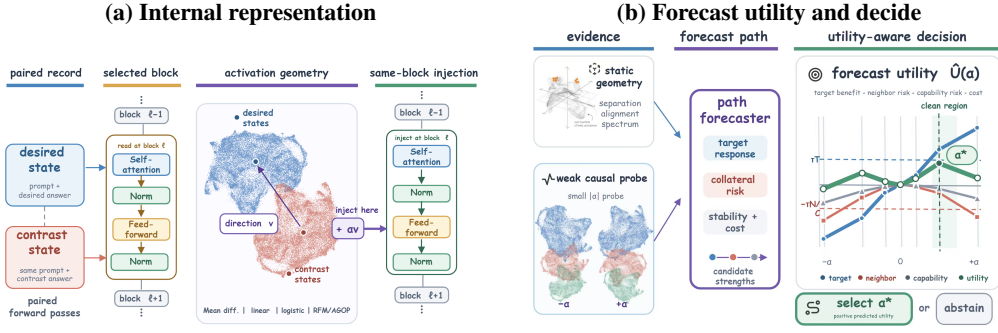


Figure 1: PML from representation to selective control. (a) Direction estimators use desired and contrast activations at a selected block, where the resulting direction is injected. (b) Static evidence and a disjoint low-dose response forecast target benefit, collateral risk, and utility over candidate coefficients for selection or abstention. Curves are schematic.

On 3,000 frozen records from nine datasets and fourteen domains, learned directions improve target and clean-path incidence over random. Static localization adds little predictive value, whereas a disjoint low-dose response dominates later-path prediction across grouped transfer. Both findings replicate across three residual-norm-matched base models (0.801–0.828 record-held-out macro AUROC). A held-out selector improves utility and reduces neighbor damage relative to a fixed intervention. The central result is that **a small causal response is the most useful forecast of later selective behavior**.

This work makes three contributions:

- We define a random-calibrated measured-grid path that jointly records target leverage, semantic-neighbor damage, capability damage, and clean measured coefficients for each record, direction, and layer.
- We separate static localization and supervised geometry from a strength-disjoint causal probe, showing that the former are weak predictive priors while the latter supplies the dominant signal across grouped and cross-model evaluation.
- We connect forecasting to action with a held-out collateral-aware policy that selects one coefficient or abstains, providing a proof of concept for reducing dense intervention evaluation.

2 RELATED WORK

Knowledge representation, localization, and editing. Transformer internals have been associated with feed-forward memories, knowledge neurons, and causally important states (Geva et al., 2021; Dai et al., 2022; Meng et al., 2022). Editing uses learned editors, external memory, or targeted parameter updates (De Cao et al., 2021; Mitchell et al., 2022a;b; Meng et al., 2023), with efficacy, generalization, and locality organized by surveys and toolkits (Yao et al., 2023; Wang et al., 2024a). Localization need not identify components that support editing or unlearning (Hase et al., 2023; Lee et al., 2025); PML tests its predictive connection to intervention outcomes.

Activation steering and control tradeoffs. Activation engineering intervenes on intermediate representations (Turner et al., 2024); contrastive activation addition, representation engineering, and affine representation surgery construct or analyze activation interventions from examples and population structure (Rimsky et al., 2024; Zou et al., 2025; Singh et al., 2024). Other work studies truthfulness components, coefficient scaling, and preference–utility tradeoffs (Li et al., 2023; Stoehr et al., 2024; Xu et al., 2026a). AxBench compares concept detection and steering methods (Wu et al., 2025), while broader evaluations expose sensitivity to instruction phrasing, prompt distribution, layer choice, and model scale (Tan et al., 2024; Da Silva et al., 2025; Goyal & Daumé III, 2026). These studies motivate joint leverage and side-effect measurement; PML additionally asks whether localization evidence forecasts their coexistence for each record–direction–layer path.

Evaluation and prediction of steerability. Steering evaluations call for task-matched contexts, likelihood-aware metrics, and baselines (Pres et al., 2024), and test control across behavioral granularities (Xu et al., 2026b). Most closely related, Fan et al. (2026) predict under-, successful, or oversteering from early decoding dynamics and rank strengths to reduce rollouts. PML instead predicts a record-direction-layer path on a pre-specified coefficient grid, calibrates target, semantic-neighbor, and capability events against random directions, and optimizes collateral-aware margin utility. Its weak feature is measured at a coefficient excluded from the stronger-coefficient labels, after which a policy chooses one coefficient or abstains. SteerBoost targets efficient generation-level concept alignment; PML tests whether localization and a fixed low-dose diagnostic forecast selective margin outcomes.

Supervised geometry as a diagnostic. Recursive feature machines use the average gradient outer product to learn task-adapted geometry (Radhakrishnan et al., 2024). PML uses its leading direction, spectral concentration, and alignment as candidate signals, while explicitly testing rather than assuming that supervised sensitivity implies selective control.

3 PREDICTIVE MEMORY LOCALIZATION

PML connects a localized residual representation to a measured strength-indexed intervention path and a held-out decision (Figure 1). Here, “memory localization” denotes a record-conditioned internal signal associated with an answer relation, not a claim that the relation resides in one unique physical component.

3.1 RECORDS, PROBES, AND INTERVENTIONS

Record i contains target prompts $\mathcal{T}_i$, semantic-neighbor prompts $\mathcal{N}_i$, and general-capability prompts $\mathcal{C}_i$. The first set expresses the behavior to suppress or enhance; the latter two test whether nearby knowledge or unrelated abilities are preserved. For prompt x with desired answer y^+ and contrast y^- , the answer margin is

$$m(x) = \log p(y^+ | x) - \log p(y^- | x). \quad (1)$$

All outcomes are changes from the unperturbed margin, making paths comparable across records with different baseline confidence.

At layer ℓ , method s constructs a unit-norm record-specific direction $\mathbf{v}_{i\ell s}$. Intervention strength α modifies the residual activation as

$$\mathbf{h}'_\ell = \mathbf{h}_\ell + \alpha \mathbf{v}_{i\ell s}. \quad (2)$$

Negative and positive coefficients test suppression and enhancement with the same direction. We write $\Delta_i^T(\alpha)$ for signed target-margin movement and $\Delta_i^N(\alpha)$, $\Delta_i^C(\alpha)$ for neighbor and capability movement; a sufficiently negative collateral change is damage.

3.2 RANDOM-CALIBRATED INTERVENTION PATHS

Target, neighbor, and capability responses have different null scales, so random directions define separate 95th-percentile thresholds τ_T , τ_N , and τ_C . A target effect crosses τ_T in the intended direction, while neighbor or capability damage crosses the corresponding negative threshold. A *clean strength* produces a target effect while crossing neither damage threshold at that same coefficient.

Across an ordered, finite strength set, these events form a measured-grid intervention path. Target and collateral onset identify the first observed crossing, and adjacent clean strengths form an observed clean region. We use these grid-defined events as the path estimand. The confirmatory prediction task focuses on four directly measured events at held-out stronger coefficients: *Target-any*, semantic-neighbor damage, capability damage, and *Clean-any*. Percentages count record-method-layer paths rather than prompts or individual strengths.

3.3 PREDICTION TASK

PML forecasts later path outcomes from progressively stronger evidence. Baseline belief B and metadata M are augmented with static localization features L and, for RFM, geometry features G .

Low-dose response features R measure target and collateral movement at $\alpha = \pm 0.1$. Because R requires intervention, it is a cheap dynamic diagnostic rather than static localization. The central comparisons test whether L or G improves on $B + M$, and whether the strength-disjoint response R forecasts outcomes at $|\alpha| \in \{0.25, 0.5\}$. A downstream policy then uses these forecasts to select a coefficient or abstain.

4 SIGNALS AND PREDICTIVE MODELS

PML evaluates a common path-prediction and decision pipeline over several direction families, separating the quality of a direction from the evidence used to forecast its later behavior.

4.1 DIRECTION FAMILIES

Random control. A seeded unit-norm Gaussian direction defines response thresholds and the paired null baseline. The logged `matched_norm_random` entry is duplicated this direction and is retained in the audit record.

Mean difference. For positive and contrast activation sets $\mathcal{H}^+$ and $\mathcal{H}^-$, we use

$$\mathbf{v}_{\text{mean}} = \frac{\boldsymbol{\mu}^+ - \boldsymbol{\mu}^-}{\|\boldsymbol{\mu}^+ - \boldsymbol{\mu}^-\|_2}, \quad (3)$$

the multi-example analogue of activation addition (Turner et al., 2024; Rimsky et al., 2024).

Linear and logistic probes. Normalized classifier weights provide supervised discriminative directions without nonlinear feature learning.

RFM/AGOP direction. A recursive feature machine estimates task-adapted geometry through the average gradient outer product (Radhakrishnan et al., 2024). Its leading eigenvector provides a low-rank direction, while spectrum, concentration, and alignment statistics become geometry features. A top- k boundary study tests whether broader supervised subspaces trade selectivity for leverage.

All directions are fitted independently per record and layer and injected at the selected residual block during evaluation. Detailed activation extraction, fitting hyperparameters, inference settings, and continuation scoring are included in the released protocol.

4.2 PREDICTIVE EVIDENCE AND GROUPED EVALUATION

Static localization L summarizes class separation, projection, saliency, and direction agreement; RFM geometry G summarizes spectral concentration and alignment. These feature groups are nested with baseline belief B , metadata M , and the low-dose response R , allowing the contribution of static localization to be measured separately from the observed response.

We fit class-balanced logistic regression and random forests for target, damage, and clean outcomes. Five-fold evaluation groups all paths from the same record, dataset, or domain, preventing related trajectories from crossing a train-test boundary. We report prevalence, AUROC, and average precision in the main paper; additional classification and calibration metrics are in the released artifacts.

4.3 RISK-AWARE STRENGTH SELECTION

For each nonzero candidate coefficient, the selector predicts clean-effect probability and continuous utility. With $q(\alpha) \in \{-1, +1\}$ denoting the requested suppression or enhancement sign, measured utility is

$$u_i(\alpha) = q(\alpha)\Delta_i^T(\alpha) - \max\{0, -\Delta_i^N(\alpha)\} - \max\{0, -\Delta_i^C(\alpha)\}. \quad (4)$$

Target movement is rewarded and collateral margin decreases receive unit penalties. The decision score is

$$\hat{u}_i(\alpha) + 0.1\hat{p}_i(\text{clean} \mid \alpha) - 0.01|\alpha|. \quad (5)$$

The policy selects the highest-scoring coefficient and abstains when its score is nonpositive. This utility is defined on teacher-forced answer-margin changes. We evaluate it against both no intervention, whose utility is zero by construction, and a train-tuned fixed coefficient.

5 EXPERIMENTS

We organize the evidence around three questions: whether learned directions create selective intervention paths, which internal signals forecast those paths, and whether those forecasts improve strength decisions. We report the frozen confirmatory study here; the Supplementary Material adds protocol details, uncertainty estimates, and analyses not shown in the main paper.

5.1 FROZEN MULTIDOMAIN STUDY

The benchmark contains 3,000 records from nine public sources: MMLU-Pro, MMLU-Redux 2.0, AI2 ARC, OpenBookQA, SciQ, LiveBench reasoning and math, HellaSwag, and QASC (Wang et al., 2024b; Gema et al., 2025; Clark et al., 2018; Mihaylov et al., 2018; Welbl et al., 2017; White et al., 2025; Zellers et al., 2019; Khot et al., 2020). The collection spans fourteen academic, scientific, commonsense, mathematical, and reasoning domains. A schema-constrained generation step converts each source item into disjoint direction-fitting statements, three record-specific target probes, and three semantic-neighbor probes; four globally balanced capability probes are assigned per record. The worked example below traces one source item from fitting evidence to evaluation probes and its resulting path label. Dataset composition, validation, and additional examples appear in the Supplementary Material.

Worked example. One frozen record from construction to path label; direction-fitting statements and evaluation probes are disjoint.

Object	Frozen example
Source	MMLU-Redux 2.0 chemistry: Suppose that the ^{13}C nuclei in a molecule in a 600 MHz spectrometer can be 100% polarized ($p = 1$). If $T_1 = 5.0$ s, how long does it take for p to reach a value equal to twice the thermal equilibrium polarization at 298 K?
Fit evidence	Positive statement: The polarization reaches twice the thermal equilibrium value in 72.0 seconds. Contrast statement: The polarization reaches twice the thermal equilibrium value in 56.6 seconds.
Target probe	Prompt: After full polarization, how long until it reaches twice thermal equilibrium?; candidate answers: 72.0 s vs. 56.6 s.
Neighbor probe	Prompt: The spin-lattice relaxation time T_1 for ^{13}C in this experiment is; candidate answers: 5.0 s vs. 10.0 s.
Capability probe	Prompt: An object in motion tends to stay in motion unless acted upon by an external force. This is; candidate answers: Newton’s first law vs. Newton’s second law.
Measured path	Mean difference, layer 11: suppression onset -0.25 ; positive neighbor onset 0.5 ; no enhancement or capability onset.
Labels	Target=1, N-dmg=1, C-dmg=0, Clean suppression=1, Clean=1.

We evaluate Qwen3-1.7B-Base (Yang et al., 2025) at two middle-depth Transformer blocks, reported as blocks 7 and 11 by the model implementation and selected using a 500-record layer-selection subset (Figure 2). We compare five direction constructions over a signed coefficient sweep. Random directions calibrate target and collateral thresholds, while the weak response at $|\alpha| = 0.1$ is disjoint from the stronger coefficients used to define confirmatory outcomes. Table 1 defines the reported path rates, and prediction folds hold out complete records, datasets, or domains.

For cross-model confirmation, the same frozen 500-record subset is evaluated on Qwen3-1.7B, Qwen3.5-2B-Base (Qwen Team, 2026), and Ministral-3-3B-Base (Liu et al., 2026). We align relative intervention budget $\|\alpha v\|_2 / \|h\|_2$ using

$$s_{m,\ell} = \frac{\text{RMS}(h_{m,\ell})}{\text{RMS}(h_{\text{ref},\ell})} \sqrt{\frac{d_m}{d_{\text{ref}}}}, \quad \alpha_{m,\ell} = s_{m,\ell} \alpha_{\text{ref}}. \quad (6)$$

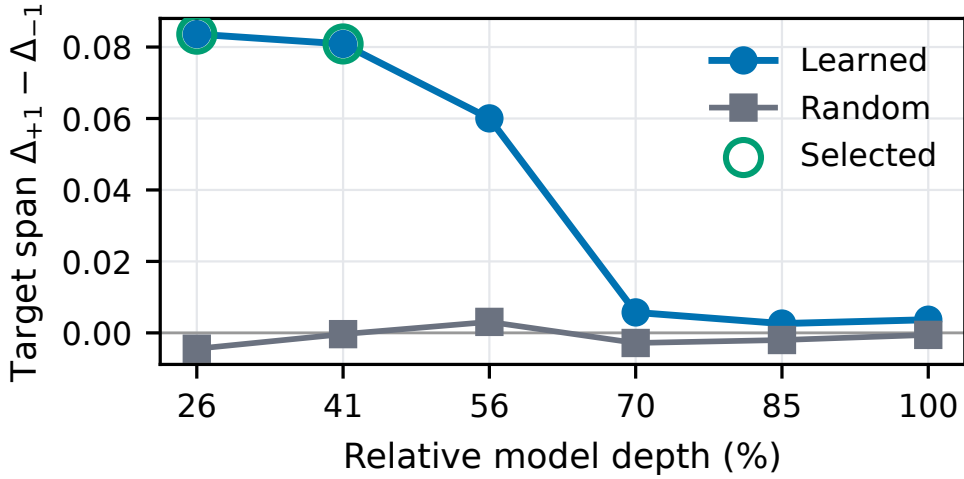


Figure 2: Layer selection on a 500-record subset. Target-margin span across relative depth identifies blocks 7 and 11 as the strongest distinct middle-depth blocks; random spans remain near zero.

Scales are fixed on a disjoint 100-record calibration set, after which each model recalibrates its random-response thresholds on the formal cohort. The confirmation retains random, mean-difference, logistic, and RFM/AGOP directions at two pre-specified blocks per model. We omit linear because it is not strongest at either primary-study block, which preserves a symmetric comparison across architectures.

5.2 LEARNED DIRECTIONS IMPROVE SELECTIVE PATHS

Table 1 reports both the powered 3,000-record outcome estimate and the three pre-specified 500-record confirmations. Its entries are path-incidence percentages; ΔT and ΔC are percentage-point differences from the within-model random control, not AUROC. In the primary study, layer-7 RFM/AGOP reaches **13.1% Target and 12.3% Clean**, the strongest selective-path result. More broadly, learned directions improve target leverage and clean-path incidence over random controls. The strongest construction varies by block.

Record-paired bootstrap intervals confirm the learned-over-random Target and Clean gains for the strongest primary-study directions. Collateral incidence varies by model and layer; full intervals are reported in the Supplementary Material.

The residual-norm-matched confirmations preserve the same qualitative pattern at model-dependent magnitudes. Figure 3 shows that learned directions generally increase target leverage, while collateral incidence varies across models and layers. The Qwen3-1.7B subset also closely tracks the powered estimate, separating cohort variation from the cross-model scale alignment.

5.3 LOW-DOSE RESPONSES FORECAST LATER OUTCOMES

Prediction is the central test of PML. Static localization is a weak prior: Figure 4 establishes a consistent diagnostic hierarchy. Static localization L adds little beyond base and metadata features, while supervised geometry G offers a similarly limited gain. In contrast, the strength-disjoint weak response R produces the dominant gain across outcomes and models. The Supplementary Material further separates this gain from ordinary scalar weak-to-strong correlation. Positive-label prevalence is 8–11%, so the reported tables include AP with AUROC. For Target-any and Clean-any, a multi-variate R -only random forest substantially improves on a single signed response, with the complete predictor adding a smaller final gain. Final macro AUROC remains around **0.80–0.85** under record-, dataset-, and domain-held-out evaluation. Removing all 500 records used for layer selection leaves the four principal full-predictor AUROCs within 0.01 of the 3,000-record estimates. The same pre-

Table 1: Random-calibrated path incidence (%) for the 3,000-record primary study and three 500-record confirmations. N/C-dmg. are collateral damage; Δ columns are learned-minus-random points. Bold marks each block’s strongest learned result.

Layer	Direction	Supp.	Enh.	Target	N-dmg.	C-dmg.	Cl. supp.	Cl. enh.	Clean	ΔT	ΔC
A. Qwen3-1.7B-Base ($n = 3,000$, primary scale, $\tau_T = 0.175$)											
7	Random	5.1	4.7	9.5	8.2	8.2	4.7	4.3	8.9	–	–
7	Mean diff.	6.1	7.0	12.8	7.9	8.2	5.6	6.7	12.0	+3.3	+3.2
7	Logistic	6.7	6.1	12.4	9.1	9.0	6.3	5.7	11.6	+2.8	+2.8
7	Linear	5.8	6.2	11.6	8.8	8.4	5.2	6.0	10.8	+2.1	+2.0
7	RFM/AGOP	7.1	6.4	13.1	8.5	9.0	6.5	6.1	12.3	+3.6	+3.4
11	Random	4.7	4.2	8.8	7.0	7.1	4.5	3.8	8.3	–	–
11	Mean diff.	6.2	5.3	11.3	7.8	7.1	5.6	5.1	10.6	+2.5	+2.3
11	Logistic	5.7	5.7	11.2	7.4	7.1	5.4	5.4	10.6	+2.4	+2.3
11	Linear	5.3	5.1	10.2	6.8	7.1	5.0	4.9	9.7	+1.4	+1.4
11	RFM/AGOP	5.1	5.3	10.2	7.5	7.2	4.8	5.1	9.7	+1.4	+1.5
B. Qwen3-1.7B-Base ($n = 500$, residual-norm matched, $\tau_T = 0.177$)											
7	Random	4.4	4.8	9.0	7.4	9.2	4.0	4.0	7.8	–	–
7	Mean diff.	5.2	7.4	12.6	8.2	10.2	4.4	7.2	11.6	+3.6	+3.8
7	Logistic	6.8	6.2	12.8	8.0	8.6	6.6	5.6	12.0	+3.8	+4.2
7	RFM/AGOP	6.2	7.8	13.2	9.4	9.6	5.6	7.2	12.4	+4.2	+4.6
11	Random	4.8	4.8	9.6	7.6	9.0	4.8	4.4	9.2	–	–
11	Mean diff.	6.0	4.6	10.0	7.6	8.0	5.0	4.2	8.6	+0.4	-0.6
11	Logistic	5.8	6.4	12.2	7.8	7.4	5.6	5.8	11.4	+2.6	+2.2
11	RFM/AGOP	4.2	5.8	9.8	8.2	7.4	3.8	5.2	8.8	+0.2	-0.4
C. Qwen3.5-2B-Base ($n = 500$, residual-norm matched, $\tau_T = 0.111$)											
6	Random	4.6	4.0	8.6	7.6	8.2	3.6	3.6	7.2	–	–
6	Mean diff.	16.6	17.2	30.2	9.6	8.8	14.6	16.0	28.2	+21.6	+21.0
6	Logistic	18.8	16.0	28.4	11.2	5.6	17.0	15.2	26.8	+19.8	+19.6
6	RFM/AGOP	18.8	17.6	32.0	10.6	9.0	17.8	17.4	31.2	+23.4	+24.0
9	Random	4.6	4.8	9.4	6.8	5.4	4.2	4.4	8.6	–	–
9	Mean diff.	9.2	7.2	15.6	9.6	5.2	8.8	6.8	15.0	+6.2	+6.4
9	Logistic	7.8	8.8	16.2	8.6	6.0	7.0	8.0	14.6	+6.8	+6.0
9	RFM/AGOP	8.0	7.4	15.0	9.0	5.8	7.4	6.6	13.8	+5.6	+5.2
D. Ministral-3-3B-Base ($n = 500$, residual-norm matched, $\tau_T = 0.083$)											
6	Random	6.2	5.0	10.6	7.8	9.4	5.6	4.6	9.8	–	–
6	Mean diff.	11.2	11.2	21.2	10.6	9.8	10.4	10.6	19.8	+10.6	+10.0
6	Logistic	10.0	12.2	21.4	11.2	8.8	9.8	11.6	20.6	+10.8	+10.8
6	RFM/AGOP	11.0	9.8	19.4	11.2	11.4	10.4	9.2	18.2	+8.8	+8.4
10	Random	3.6	4.6	7.8	7.2	8.8	3.2	4.6	7.4	–	–
10	Mean diff.	9.0	8.0	16.4	8.4	6.4	8.8	7.4	15.6	+8.6	+8.2
10	Logistic	9.6	8.6	17.8	8.8	9.2	9.0	8.2	16.8	+10.0	+9.4
10	RFM/AGOP	8.6	8.2	16.0	7.6	8.0	8.0	7.4	14.8	+8.2	+7.4

dictive ordering therefore holds after removing the layer-selection records: a structured low-dose causal response is more informative about later paths than static localization geometry alone.

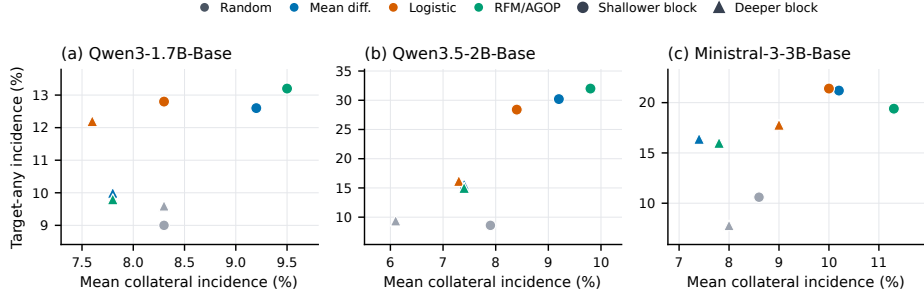


Figure 3: Target leverage and collateral incidence on the common 500-record cohort. Each panel shows one base model; colors identify direction construction and marker shapes identify the shallower or deeper pre-specified block. The vertical axis is Target-any path incidence, while the horizontal axis averages semantic-neighbor and capability-damage incidence. Each model has two pre-specified block evaluations, with panel-specific axes that keep within-model trade-offs visible.

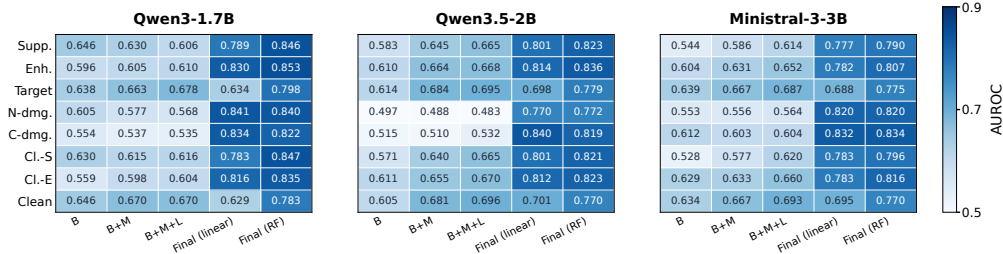


Figure 4: Cross-model path prediction. Per-outcome record-held-out AUROC on the common 500-record cohort. The first three random-forest columns cumulatively add base margins and record descriptors (B), method/block metadata (M), and static localization (L). The final two columns use the complete $B+M+L+R$ features with a linear predictor or random forest, providing a predictor-class ablation. The weak response produces the dominant feature gain under both predictors, while the nonlinear model gives the strongest final performance.

5.4 FORECASTS IMPROVE STRENGTH DECISIONS

We train candidate-strength models from complementary dense and sparse trajectories, then evaluate on disjoint dense-grid records. Before a candidate outcome is revealed, each policy selects one coefficient or abstains. The Supplementary Material gives the full 100-record proof-of-concept table. Relative to a train-tuned fixed coefficient, utility improves by **0.055** and **0.034**, mainly as neighbor damage falls from 6.0% to 1.8% and 5.2% to 2.2%. Suppression exceeds no intervention, while enhancement shows a positive but uncertain gain. The policy averages 2.55/2.61 coefficient evaluations—two weak probes plus a final action—versus 26 for a dense scan; shared direction fitting is excluded. Weight sensitivity and full outcomes are supplementary.

5.5 ENDPOINT SCOPE

Free-generation stress tests are reported in the Supplementary Material. They show that margin movement can alter text, while wrong-to-right correction is less frequent. PML therefore evaluates teacher-forced, margin-level intervention paths.

6 DISCUSSION

Measured paths separate leverage from selectivity. Learned directions increase Target and Clean incidence, while their neighbor- and capability-damage differences remain statistically un-

resolved. A direction can therefore create more usable measured coefficients while its collateral profile varies by layer. Layer 7 offers greater leverage together with more collateral movement than layer 11. The measured clean regions show where leverage and selectivity coincide: most contain one measured clean coefficient, which motivates multi-strength evaluation alongside the primary prediction labels.

Budget matching preserves the diagnostic hierarchy. Residual-norm matching aligns relative intervention magnitude, not response rates: Qwen3.5 shows the largest gains and Ministral is intermediate. All three models nevertheless reproduce more learned clean paths and a much larger predictive contribution from low-dose response than static localization.

A small causal response is more actionable than static localization. Supervised geometry is valuable for constructing high-leverage directions and describing their concentration and alignment, while these static descriptors add little forecasting power by themselves. A strength-disjoint low-dose response strongly predicts later outcomes across record, dataset, and domain transfer. Localization becomes actionable when it is paired with a cheap causal measurement of the specific path. PML therefore complements concept detection and average steering evaluation (Wu et al., 2025; Da Silva et al., 2025; Goyal & Daumé III, 2026) by testing whether an internal signal supports effective and selective margin movement at a chosen strength.

The weak, outcome-specific transfer of activation-path features to ROME defines the scope of the result. A path can diagnose a fragile or promising activation intervention, while persistent parameter editing remains a separate problem (Hase et al., 2023).

7 LIMITATIONS

The primary study uses Qwen3-1.7B and two blocks selected on a 500-record subset. Width-corrected confirmations add Qwen3.5-2B and Ministral-3-3B at two pre-specified aligned blocks, supporting relative-depth and predictor-order claims rather than global layer optimality or absolute cross-architecture incidence comparisons. The Qwen3-1.7B subset differs slightly from the powered estimate because of sampling and threshold recalibration.

Primary outcomes are teacher-forced margins at two held-out coefficients; finite-grid topology and free-generation stress tests provide complementary scope. Schema-constrained probes receive expert audit and exclusion sensitivity, while independently authored validation remains future work. Frozen random-control percentiles and the declared utility define the within-study decision rule; deployment transfer requires recalibration.

The 100-record selector excludes shared fitting and feature-extraction costs, and neighbor and capability probes cover selected downstream effects. The small ROME transfer study provides an initial comparison with persistent editing.

8 CONCLUSION

Predictive Memory Localization forecasts target, neighbor, capability, and clean margin outcomes on a measured intervention grid. In the 3,000-record primary study and 500-record confirmations across three base models, learned directions improve target and clean incidence, while the strength-disjoint low-dose response dominates static localization for later-outcome prediction. A held-out selector improves utility over a fixed coefficient, reduces neighbor damage, and replaces a dense scan with weak probes plus action or abstention. PML therefore turns localization into a falsifiable forecast and risk-aware decision: representation evidence identifies leverage, while a small causal response reveals collateral risk before stronger intervention.

AI USE STATEMENT

Generative AI tools assisted with coding, manuscript editing, and LaTeX formatting. Benchmark preparation used schema-constrained generation of direction-fitting statements and evaluation probes, as described in Section 5.1 and the supplementary benchmark-construction details. The authors remain responsible for the methods, evidence, citations, and final manuscript.

REPRODUCIBILITY STATEMENT

The predictive task, direction families, grouped evaluation, and selector utility are specified in Sections 3–5. Supplementary Sections S1–S3 give benchmark construction, inference settings, and the cross-model protocol; Sections S4–S7 provide bootstrap, prediction, strength-selection, and end-point details. These include computational-environment information and the logged cohort used for the primary prediction estimates.

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